🧠 Livestock Health Intelligence System (LHIS)

📄 Technical Report (Final — Developer + System Design Level)

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1. 🧠 Problem Interpretation

Livestock health monitoring today is **reactive**:

- Farmers detect illness after visible symptoms
- No continuous tracking
- No integration of multiple health signals
- No personalized animal-level insights

This results in:

- Reduced productivity
- Delayed treatment
- Financial loss

🎯 Objective

To build a system that:

- Continuously tracks livestock health
- Detects early risk signals
- Explains causes
- Guides farmer decisions

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2. 🏗️ System Overview

🔧 Core System Design

```
User (Farmer)
    ↓
Frontend Dashboard
    ↓
API Layer (FastAPI)
    ↓
Core Processing Engine
    ↓
Risk + Insights + Simulation
```

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3. 🔄 End-to-End Workflow

```
User Input
  ↓
Validation
  ↓
Feature Engineering
  ↓
Context Detection (Animal State)
  ↓
Risk Engine (Rule + ML)
  ↓
Explainability Layer
  ↓
Response Generation
  ↓
Simulation (optional)
```

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4. 📥 Input Workflow

```
Farmer opens app
  ↓
Select Animal
  ↓
Enter daily data:
  - Milk (optional)
  - Feed
  - Activity
  - Temperature
  ↓
Submit
```

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5. 🧪 Validation Flow

```
Input Data
  ↓
Check ranges:
  milk ≥ 0
  temp (35–42°C)
  ↓
Check missing values
  ↓
Valid → proceed
Invalid → error
```

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6. 🧠 Feature Engineering Workflow

```
Raw Data
  ↓
Derived Signals:
  milk_drop %
  activity_score
  feed_score
  temp_risk
  ↓
Feature Vector
```

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7. 🧠 Context-Aware Logic (CRITICAL)

```
Check Animal State:
  ↓
IF lactating:
  use milk + all signals
ELSE (dry):
  ignore milk
  use behavior + health
```

👉 Ensures system works in all lifecycle stages

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8. 🤖 Risk Engine Workflow

🔹 Hybrid System

Feature Vector
↓
Rule Engine
↓
ML Model
↓
Voice Score (optional)
↓
Risk Fusion Engine
↓
Final Risk Score

🔹 Rule Logic Example

IF milk_drop > 15% AND activity_low
→ HIGH RISK

🔹 ML Logic

model.predict(features)
→ probability
→ risk score (0–100)

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9. 🔍 Explainability Workflow

Model Output
↓
Feature Importance
↓
Top Factors:
- Milk drop
- Activity low
↓

Generate explanation

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10. 🔮 Simulation Workflow (USP)

User modifies input
↓
Recompute features
↓
Re-run model
↓
Compare old vs new
↓
Return delta

Output Example

Before: 70
After: 52
Improvement: -18

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11. 🔊 Voice Module Workflow (Optional)

Audio Input
↓
Feature Extraction (pitch, energy)
↓
Classification:
normal / stress
↓
voice_score
↓
Added to risk engine (low weight)

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12. 📊 Trend Tracking Workflow

Daily data

↓
Store history
↓
Analyze trend:
milk ↓ over days
activity ↓
↓
Trend alert

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13. 📤 Output Workflow

Risk Score
↓
Risk Level (LOW/MED/HIGH)
↓
Reasons
↓
Suggested Actions

Output Example

Animal: Cow #12

Risk: HIGH (72)

Reasons:
- Milk decreased
- Activity low

Action:
- Improve feed
- Monitor closely

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14. 🏗️ System Architecture (Detailed)

Frontend (Next.js)
↓
API Layer (FastAPI)
↓

```
┌──────────────┬──────────────┬──────────────┬──────────────┐
│ Validation   │ Feature Eng  │ Context Logic│              │
└──────────────┴──────────────┴──────────────┴──────────────┘
    ▼            ▼            ▼
┌──────────────┐ ┌──────────────┐ ┌──────────────┐
│ Rule Engine  │ │ ML Model     │ │ Voice Module │
└──────────────┘ └──────────────┘ └──────────────┘
      \            |            /
       \           |           /
        ▼          ▼          ▼
        [Risk Fusion Engine]
              ↓
        [Explainability]
              ↓
        [Simulation Engine]
              ↓
        [Response]
```

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15. 🧪 ML Training Workflow

Dataset
↓
Cleaning
↓
Encoding
↓
Normalization
↓
Feature Engineering
↓
Train/Test Split (80/20)
↓
Model Training (LR / RF)
↓
Evaluation
↓
Model Save

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16. ⚙️ Tech Stack

- Frontend: Next.js

- Backend: FastAPI
- ML: scikit-learn
- Audio: Librosa
- Storage: SQLite

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17. ⚡ Performance Requirements

- API response < 2 sec
- Simulation < 1.5 sec
- Lightweight inference

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18. 🔐 Error Handling

Invalid input → 422
Missing data → 400
System error → 500

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19. 💡 Key Innovations

- Context-aware system (lifecycle-based)
- Multi-signal risk detection
- Simulation-driven insights
- Explainable outputs

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20. ⚠️ Limitations

- Manual data input
- No clinical validation
- Limited dataset

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21. 🔮 Future Scope

- IoT integration
- Real-time monitoring
- Multi-animal analytics
- Mobile deployment

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22. 🧠 Conclusion

LHIS transforms livestock health management from:

Reactive → Proactive
Static → Interactive
Guess-based → Data-driven